

FEATURED PROJECT AND BUSINESS IMPACT

As a researcher and research team leader in MNC, the first priority is still to demonstrate the value to the company. Therefore, this page selects a few representative projects to address this question.

AI + Manufacturing Projects

Computer Vision for PCBA Assisted Visual Inspection [24][36][37], 2022

Project Overview: Leveraged deep learning networks with contrastive learning to enhance model generalization. Post-processing operators were designed based on insights from veteran experts, ensuring strong interpretability across the entire solution.

Key Achievements: Deployed to mass production in June 2021; continues to serve all company PCBA production lines. Achieved zero missed detections with less than 10% false positive rate. Reduced manual visual inspection workload by over 92% at the Shanghai plant alone. Won SAIC Innovation Awards twice in 2022 and 2025, reported by Shanghai Television as well.

Smart Sensor Projects

Automatic annotation algorithm for arrhythmia [15],[18], 2017-2018

Project Overview: My first AI project and publication designed for GE diagnostic cardiology (DCAR) portfolio. Multichannel signals are transferred into frequency spectrum. Then a latent representation are learned by VAE.

Key Achievements: Integrated into DCAR product.

AI + Contrl Projects

Battery Balancing Algorithm for BMS [33][34][39], 2023

Project Overview: Designed for passive balancing scenarios, the objective was to identify optimal strategies that enable the controller to guide the circuit—avoiding thermal protection triggers while minimizing balancing time. including:

- Built a mechanism model using Simscape
- Calibrated hyperparameters with physical measurement data
- Developed a customized NeuralODE to fit the mechanism model, creating an AI surrogate model
- Constructed a Gym environment
- Learned balancing strategies via reinforcement learning
- Analyzed RL-derived strategies and simplified them into robust heuristic strategies

Key Achievements: Integrated into BMS product in 2024, improving balancing efficiency by over 10% compared to conventional strategies.

AI + Supply Chain Projects

APAS Algorithm [32], 2020 & 2025

Project Overview: Utilized mixed-integer programming (MIP) and constraint programming (CP) from operations research. Given order demands, production line manufacturability constraints, and tooling cart constraints, the system determines optimal capacity planning and production scheduling.

Key Achievements: Implemented in two phases.

- Phase 1 (April 2020): Capacity planning module launched in the IE system, using the CBC solver.
- Phase 2 (2025): Production scheduling algorithm launched in September, using IBM CPLEX. Currently operational at the Liuzhou factory.

Algorithm vs. manual capacity planning: 5.2% labor hour savings, 2.7% equipment time savings.

Algorithm vs. manual production scheduling (at 100% capacity utilization): 6% reduction in order lead time, 1.8% fewer tooling cart changeovers.

Agents Projects

Agents for R&D Efficiency [47][48][49][50][51][52][54], 2023 –Present

Project Overview: By combining large language models, context engineering, ontology and knowledge curation, MAS reasoning, and tool orchestration, the initiative comprehensively empowers the company's R&D processes, including two major R&D portals.

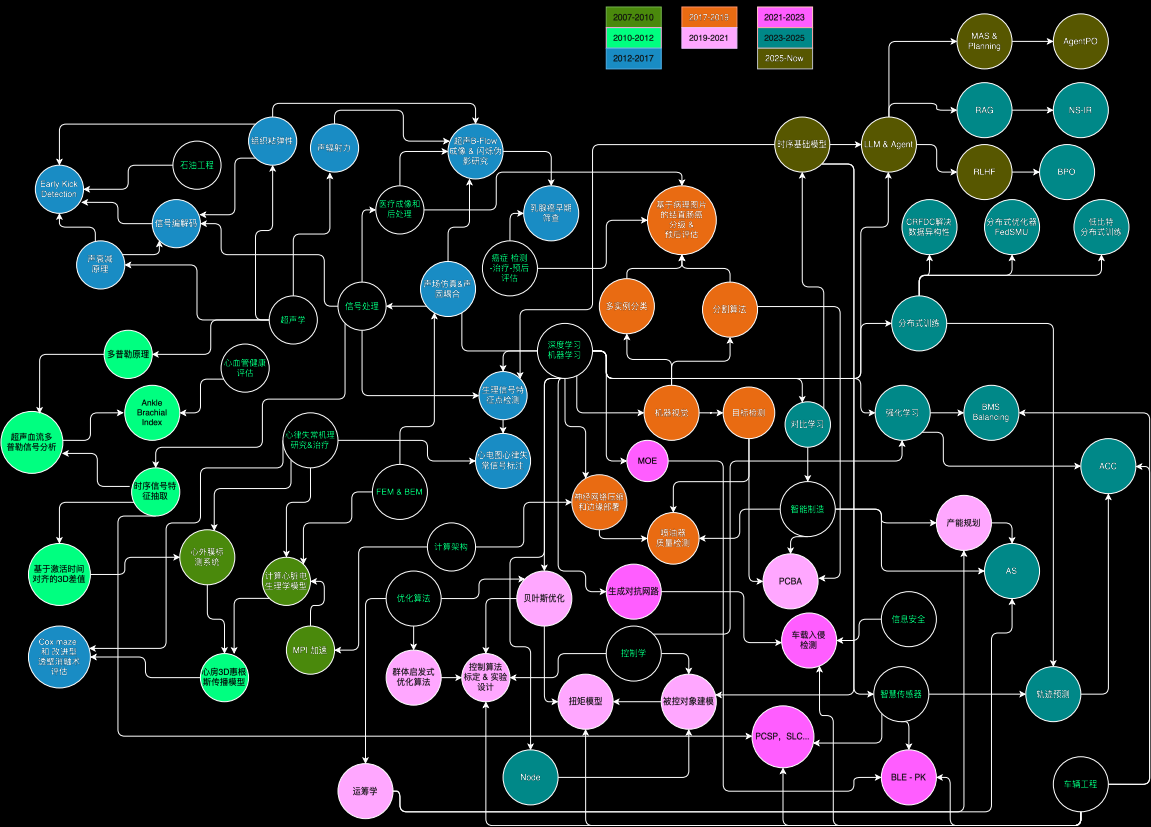
Key Achievements: Improvements in the system development ASPICE process model: System Requirements Analysis (+5%), System Architecture Design (+11%), System Integration (+8%), ASW Software (+3%–9%), BSW Software (+2%–18%). Skill duplicate loading rate: 64%. Portal bots serve 15% of employees monthly. The self-developed PDF structured parsing tool averages 1,000+ daily uses. And copilot product called Ulystudio in USP 3.0.

RESEARCH OVERVIEW

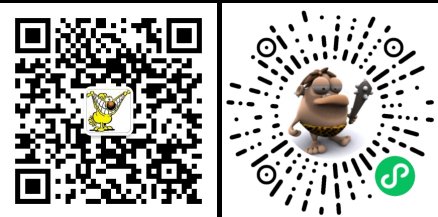
It's believed that a CV is the snapshot of human life, thus this page is prepared to reveal the academic life of Weijia, which can be broadly divided into the following stages:

- Stage I: explore physical world with measurement instruments + mechanism simulation.
- 2007–2010: Specialized in medical electrophysiology research, investigating arrhythmia detection and mechanisms through hardware system design, software development, time-series signal processing algorithms, and computer simulation. Simulations encompassed cell transmembrane potential models, Huygens propagation models, and electric field simulation using the boundary element method.
- 2010–2017: Expanded research to medical, oil & gas, and other sectors, applying signal processing and computer simulation to address industry-specific challenges. Signal processing evolved from 1D time-series signals to 2D ultrasound images, while simulations extended from electric fields to acoustic fields, fluid-structure coupling, acoustic radiation computation, medium attenuation, and ultrasound imaging.
- Stage II: polish tool and empower mechanism system with edge-AI model + data driven methodology.
- 2017–2019: With TensorFlow's emergence, AI was adopted for electrophysiological signal processing starting in 2016. Then focused on AI-driven analysis of medical pathology images, including instance segmentation, cancer grading based on segmentation results, and prognosis assessment from histopathological images.
- 2019–Present: Leading a team dedicated to solving critical automotive industry challenges with AI, thus provides direct product augmentation. e.g. computer vision for industrial manufacturing, optimization algorithms for accelerated calibration, AI+information security, smart sensor development, AI+control systems, plant modeling, and federated training.
- Stage III: leverage large model to accelerate exploration and exploitation.
- 2023–Present: With the advent of large models, leading team to build AI agents and company-wide ChatBots. Research contributions focus on information retrieval, large model inference, ontology, knowledge engineering and self-evolving agents.

Here's a diagram briefly shows the developing of research topic (A interactive version please goto this link):



Moreover, if you need details, please follow Weijia's Wechat Official Account (ALF & SAM) and use AskMe Bot in service.



FEATURED PUBLICATIONS

30+ papers, 60+ patents, those with following symbols are recommended:

(♣) Software implementation and algorithm; (♠) Signal Processing, deep/machine learning and computer architecture, AI for control, plant modeling; (♥) Physics and computational modeling; (♦) Optimization algorithm and operational research.

1. Design and Implementation of a New System for Whole-Atrial Epicardial Mapping

👤 Cuiwei Yang, [Weijia Lu](#), Xiaomei Wu, and Zuxiang Fang

📅 2007 📖 International Journal of Bioelectromagnetism (IJB)

🔗 [Link](#)

About: This study introduces the way to design an electronic system to records electrophysiology activity of heart. Cardiac Mapping and Cardiac Modeling.

2. A New Scheme for Observation and Interpretation Atrial Fibrillation

👤 [Weijia Lu](#), Zuxiang Fang

📅 2008 📖 in Proceedings of the 2nd International Conference on Bioinformatics and Biomedical Engineering (ICBBE)

🔗 [Link](#)

3. A Visual Expression to Show Epicardial Electrical Activity Comprehensively

👤 Tou Zhou, [Weijia Lu](#), Cuiwei Yang, and Zuxiang Fang

📅 2008 📖 in Proceedings of the 2nd International Conference on Bioinformatics and Biomedical Engineering (ICBBE)

🔗 [Link](#)

4. Dynamic Epicardial Mapping Using 3D Emulation

👤 [Weijia Lu](#), Tuo Zhou, Cuiwei Yang, and Zuxiang Fang

📅 2008 📖 in Proceedings of the International Conference on Biomedical Engineering and Informatics (ICBEI)

🔗 [Link](#)

5. Development of Epicardial Mapping System for Study Atrial Fibrillation

👤 Cuiwei Yang, [Weijia Lu](#), Tuo Zhou, Xiaomei Wu, and Zuxiang Fang

📅 2008 📖 in Proceedings of the International Conference on Biomedical Engineering and Informatics (ICBEI)

🔗 [Link](#)

6. A Method for Real-time Sampling and Smoothly Scrolling in Epicardial Mapping System (♣)

👤 [Weijia Lu](#), Cuiwei Yang, and Zuxiang Fang

📅 2009 📖 Journal of Biomedical Engineering (JBE Chinese), vol.26, pp.1102-1105

🔗 [Link](#)

About: This study introduces software design of employing DirectX to smoothly scrolling multiple signals on screen in a high speed sampling scenario. The corresponding GUI of system reported in IJB2007. Cardiac Mapping and Cardiac Modeling.

7. A Parallel Algorithm for Computer Simulation of Electrocardiogram Based on MPI (♣)

👤 Wenfeng Shen, [Weijia Lu](#), Daming Wei, Weimin Xu, Xin Zhu, and Shizhong Yuan

📅 2009 📖 in Proceedings of 8th IEEE/ACIS International Conference on Computer and Information Science (ICIS)

🔗 [Link](#)

About: This study is about software design of ECG computational simulation in HPC. A HPC version of Wei-Harumi Model. And the Wei-Harumi Model is a pioneering whole-heart computer simulation model for cardiac electrophysiology, based on Huygens' Principle Propagation and developed by Dr. Daming Wei. Cardiac Mapping and Cardiac Modeling.

8. Implementation of a Novel Interpolating Method to Epicardial Potential Mapping for Atrial Fibrillation Study (♣♣♣)

👤 [Weijia Lu](#), Cuiwei Yang, Zuxiang Fang, Xingpeng Liu, Xin Zhu, and Daming Wei

📅 2010 📖 Computers in Biology and Medicine (CBM), vol.40, pp.456-463

🔗 [Link](#)

About: An Spatio-Temporal interpolation algorithm to estimation a high resolution electrophysiology field. Traditional interpolation only consider the spatial relationship of sampling position while our method combines the spatial calculation with electrophysiology activity propagation. Designed for system reported in IJB2007. Cardiac Mapping and Cardiac Modeling.

9. A Computer Model Based on Real Anatomy for Electrophysiology Study (♥♥♥♣)

👤 [Weijia Lu](#), Daming Wei, Xin Zhu, and Wenxi Chen

📅 2011 📖 Advances in Engineering Software (AES), vol.42, pp.463-476

🔗 [Link](#)

About: An computational model buildup based on real anatomy information to simulate 12 channel ECG. A upgrading of Wei-Harumi Model by introducing real anatomy, cellular Ion Channel description and atrial propagation system. Especially for atrial arrhythmia study. And the Wei-Harumi Model is a pioneering whole-heart computer simulation model for cardiac electrophysiology, based on Huygens' Principle Propagation and developed by Dr. Daming Wei. Cardiac Mapping and Cardiac Modeling.

10. Method and Device for Detecting Occlusion/Reopening of an Artery and System for Measuring Systolic Blood Pressure (♠)

👤 Yinan Chen, [Weijia Lu](#), Jianyi Zhong, Ajay Anand, John Petruzzello

📅 2012 📖 US 10076310 B2

🔗 [Link](#)

About: Method to detecting blood pressure using pulse wave ultrasound. First achievement during my career path fulfilled in Philips Research. Handcrafted signal feature is extracted from IQ data to predict the state of artery. Medical Ultrasonic and Signal Processing.

11. Computer Simulation of Cathode Ablation for Atrial Fibrillation (♥)

👤 Xin Zhu, Di Yang, [Weijia Lu](#), Wenxi Chen, Daming Wei, Koji Fukuda, and Hiroaki Shimokawa

📅 2014 📖 in Proceedings of 14th IEEE International Conference on Computer and Information Technology (CIT)

🔗 [Link](#)

About: Simulation atrial fibrillation using computational model reported in AES2011 and created by Dr. Weijia Lu. The simulation results in this study demonstrated that non-transmural maze III ablation procedure can achieve the same effect as the transmural ablation if the positions of ablation lines are reasonable. Thus this paper serves as another clinic evidence of the efficiency of computational model reported in AES2011. Cardiac Mapping and Cardiac Modeling.

12. Method to Develop Coded Excitation for Velocimetry in Downhole Drilling (♠♠♥)

👤 [Weijia Lu](#), Ran Niu, Longtao Yuan, Xin Qu, Heng Wu, Jing Ye

 2015  in Proceedings of 15th IEEE International Conference on Computer and Information Technology (CIT)  [Link](#)

About: Encoding-decoding algorithm for pulse wave ultrasound, which can significantly improving spatial resolution without jeopardizing signal penetration. By-product of my studying related to B-Flow (JMU2017) and designed for early kick detection project, and my first publication in GE Research. Details are as following: In this paper, the phase modulation coding/decoding technology is discussed for the application in the high attenuation medium based on the simulation and the experiment. For the decoding perspective, the matched filter and the inverse filter are compared. An extra processing is introduced right after the matched filtering to restrain the range sidelobes. Furthermore, in the latter part of this paper, the factors to cause a degeneration in signal-noise ratio are quantitatively evaluated by simulation beam pattern and Doppler signal. Based on the experiments and simulation in this paper, it's found that the signal amplitude get significant boosted (30dB) while the range resolution is compatible with the span of a single pulse. It's also proved that the frequency dependent phase distortion from attenuation will cause more degeneration than Doppler effect, but even this influence can be compensated by the equalization. Industry Ultrasonic and Signal Processing.

13. Dominant Factor Analysis of B-flow Twinkling Sign with Phantom and Simulation Data (♠️♥️♥️♥️)

 Weijia Lu, Bruno Haider

 2017  Journal of Medical Ultrasonics (JMU), vol.44, pp.37-50  [Link](#)

About: A mechanistic study of twinkly phenomenon showup in B-Flow ultrasonic imaging. A study established a multiphysics computational model to describe the acoustic coupling of ultrasonic field and solid granules, then designed phantom study to validate the mechanistic theory. The agreement of simulation and phantom study is found by a post-processing algorithm in also reported in this study. The break-through achievement is acknowledged by the chief engineer of medical ultrasound BU. Medical Ultrasonic and Signal Processing.

14. Sensing Systems and Methods for Detecting Changes in Downhole Hydrocarbon and Gas Species (♠️♥️)

 Weijia Lu, Yi Liao

 2017  WO2018170838A1  [Link](#)

About: A study of composition analysis using ultrasound attenuation also for early kick detection. Industry Ultrasonic and Signal Processing.

15. Method to Annotate Arrhythmias by Deep Network (♠️♠️♠️)

 Weijia Lu, Jie Shuai, Shuyan Gu, Joel Xue

 2018  in Proceedings of 18th IEEE International Conference on Computer and Information Technology (CIT)  [Link](#)

About: Automatic annotation of arrhythmias on ECG signal for GE Diagnostic Cardiology portfolio. My first publication about deep learning and AI, my leading collaboration of GE Healthcare China and GE Healthcare US. AI for Diagnostic Cardiology.

16. New Boundary Constraint Loss to Facilitate Glands Segmentation (♠️♠️♠️)

 Weijia Lu, Jianhua Yao, Xiao Han, Haocheng Shen

 2019  Journal of Medical Imaging and Health Informatics (JMIHI)  [Link](#)

About: Segmentation of glands on pathological image, learning method by a new loss function. My first publication of using AI on medical image and fulfilled in Tencent AI Lab. AI for Pathologic Image.

17. An Attentive Pruning Method for Edge Computing (♠️)

 Yang Gao, Hao Gong, Weijia Lu, Chen Su, Zhang Ni and Qinghua Wang

 2019  in Proceedings of 20th International Conference on Machine Learning and Computing (ICMLC)  [Link](#)

About: Method to prune object detection network, first publication of UAES AI Lab after my enrollment of UAES as chief AI scientist. Embedded AI.

18. System and Method for Identifying Cardiac Arrhythmias With Deep Neural Networks (♠️♠️♠️)

 Weijia Lu, Shuyan Gu, Joel Xue, Jie Shuai, Hu Lifei

 2020  US 10869610 B2  [Link](#)

About: The corresponding patent of publication CIT2018, this patent is authored by Joel, the principle engineer of GE Diagnostic Cardiology, and the coauthor of my publication CIT2018. AI for Diagnostic Cardiology.

19. Microsatellite Instability Prediction of Uterine Corpus Endometrial Carcinoma Based on HE Histology Whole-Slide Imaging (♠️♠️♠️)

 Tongxin Wang, Weijia Lu, Fan Yang, Li Liu, Zhong-Yi Dong, Weimin Tang, Jia Chang, Wenjing Huan, Kun Huang and Jianhua Yao

 2020  in Proceedings of IEEE 17th International Symposium on Biomedical Imaging (ISBI)  [Link](#)

About: An new AI paradigm to predict MSI on pathological image. A multi-instance learning method by Dr. Weijia Lu and Tongxin, when Tongxin was working as an intern in Tencent AI Lab. AI for Pathologic Image.

20. Development and interpretation of a pathomics-based model for the prediction of microsatellite instability in Colorectal Cancer (♠️♠️♠️)

 Cao Rui, Fan Yang, Si-Cong Ma, Li Liu, Yu Zhao, Yan Li, Dehua Wu, Tongxin Wang, Weijia Lu, Wei-Jing Cai, Hong-bo Zhu, Xue-Jun Guo, Yuwen Lu, Jun-jie Kuang, Wenjing Huan, Wei-min, Tang, Kun Huang, Junzhou Huang, Jianhua Yao and Zhong-Yi Dong

 2020  Theranostics  [Link](#)

About: A collaboration research fulfilled by Tencent AI Lab and Nanfang hospital, and a systematic description of methodology reported in ISBI2020. ISBI2020 is a study fulfilled by Dr. Weijia Lu and Tongxin, and this paper is an extension organized by Dr. Fan Yang. More experiments are reported in this paper. AI for Pathologic Image.

21. Processing Methods, Devices, Equipment and Storage Media for Vehicle Data (♠️)

 Peng Liu, Weijia Lu, Bingyang Li, Hao Gong, Jie Zhuang and Tao Song

 2020  CN 112598133 B  [Link](#)

About: Optimization of sampling point selection for utilizing gaussian process regression in torque prediction. Comparing with ASCMO modeling method by ETAS Bosch, our method use only 70% of sampling point without any decreasing in prediction performance. AI for Smart Sensor.

22. Construction Method, Device and Storage Medium for Engine Exhaust System Temperature Model (🔮)

 Bingyang Li, Hao Gong, Weijia Lu, Peng Liu, Chunshan Ma, Yang Wang, Jianqiang Wang and Zhiwei Wang

 2021  CN 113239533 A  [Link](#)

About: A swarm intelligence method to optimize super-parameter map of the control algorithm. AI for smart calibration.

23. Dual Batch Size Training: An efficient MGD adaptive batch size method

 Yuhang Du, Wenfeng Shen, Baohua Liu, [Weijia Lu](#) and Hao Gong

 2021  in Proceedings of 2021 IEEE 33rd International Conference on Tools with Artificial Intelligence, ICTAI

 [Link](#)

24. Method, Device and Storage Medium of PCB Welding Defect Detection (♠♠)

 [Weijia Lu](#), Peng Liu, Bingyang Li, Chuang Liu, Wei Shen, Huan Ge, Yu Jing, Jie Zhang, Qi Wang and Yu Cao

 2021  CN 113506243 B

 [Link](#)

About: A two-stage method to predict welding defect areas and defect type based on selective welding image. Segmentation AI model as well as handcrafted feature from segmentation result are introduced in this patent. AI for manufacturing.

25. Method, Device and Storage Medium of Image Recognition for Chip Welding Defect (♠)

 Peng Liu, [Weijia Lu](#), Bingyang Li, Chuang Liu, Tong Ma and Fayu Qian

 2021  CN 113724218 B

 [Link](#)

About: Welding defect detection based on resistance welding image. AI for manufacturing.

26. Using EBGAN for Anomaly Intrusion Detection (♠♠♠)

 Yi Cui, Wenfeng Shen, Jian Zhang, [Weijia Lu](#), Chuang Liu, Lingge Sun and Sisi Chen

 2022  in Proceedings of 2022 International Joint Conference on Neural Networks, IJCNN

 [Link](#)

About: A generative adversarial network to detect intrusion on vehicle gateway. AI for Vehicle Safety.

27. Knock detection method and device for PCSP ignition strategy (♠)

 Xiaofeng Ma, [Weijia Lu](#), Gang Xi and Jianqiang Wang

 2022  CN 114781425 B

 [Link](#)

About: A new knock detection method when traditional algorithm failed on PCSP ignition strategy. This method deal with labeling noise by cross training and carefully designed hand-crafted features. AI for Smart Sensor.

28. Method for counting motor rotations (♠)

 Xiaodong Zhang, [Weijia Lu](#), Zhifei Yang and Aicang Li

 2022  CN 115800644 B

 [Link](#)

About: Hand-scrafted feature calculation for ripple counter motor.

29. Gradient-Based Meta-Learning Using Uncertainty to Weigh Loss for Few-Shot Learning

 Lin Ding, Wenfeng Shen, [Weijia Lu](#), Peng Liu and Shengbo Chen

 2023  in Proceedings of ICCECE

 [Link](#)

About: Extended from Model-Agnostic Meta-Learning (MAML), we propose two improved methods for the meta-loss part: Method 1 generates weights by comparing meta-loss differences to improve the accuracy when there are few classes, and Method 2 introduces the homoscedastic uncertainty of each task to weigh multiple losses based on the original gradient descent, as a way to enhance the generalization ability to novel classes while ensuring accuracy improvement

30. Towards Designing an Attentive Deep Trajectory Predictor Based on Bluetooth Low Energy Signal (♠♠♠♠)

 [Weijia Lu](#), Xiaofeng Ma, Xiaodong Zhang, Zhifei Yang and Qinghua Wang

 2023  in Proceedings of 57th Annual Conference on Information Sciences and Systems (CISS)

 [Link](#)

About: A small but carefully designed MOE network to predict cellphone location in a key-less entry scenario of vehicle. The deep learning network, with only 700 floating parameters, has been deployed in a ECU with 300MHz frequency and limited code segment. This network has two branches, one to predict the angle and another one for radial distance, and whole network is sparse activated. Moreover a carefully designed loss function is reported in this study to accelerate network training. AI for Smart Sensor.

31. Distributed Training Methods and Systems for Models (♠♠)

 [Weijia Lu](#), Xiaodong Zhang, Zhifei Yang, Xiaofeng Ma, Chuang Liu and Wangchen Lin

 2023  CN 116822619 B

 [Link](#)

32. A Method for Automatic Capacity Allocation (◇◇)

 Shuyu Jiang, [Weijia Lu](#), Na Li, Huan Ge and Bingyang Li

 2023  CN 116384669 A

 [Link](#)

About: Automatic production line allocation using linear programming and cbc solver. AI for Scheduling.

33. A Power Battery Balancing Controller, Balancing Control Method, and Electric Vehicle (♠♠♠♠)

 Chuang Liu, Xichun Ke, Zhifei Yang, [Weijia Lu](#), Xiaodong Zhang and Xiang Di

 2023  CN 116674432 A

 [Link](#)

About: A heuristic strategy for battery balancing. In nowadays, heterogenization of power battery cells becomes a critical factor of e-car lifespan. Certain chip has been provided to automatically initiate balancing process and ultimately ameliorate the heterogenization. But the chip will shutdown balancing once the temperature reaches a pre-set threshold. So this patent introduce a control-policy to fulfilled balancing process without trigger the temperature protection strategy. AI for Control.

34. A Reinforcement Learning-based Battery Balancing Method and Device (♠♠)

 Zhifei Yang, Xichun Ke, Chuang Liu, [Weijia Lu](#), Xiaodong Zhang and Xiang Di

 2023  CN 116767024 A

 [Link](#)

About: The reinformant learning verion of CN 116674432 A. Moreover this patent reports the method to establish the digital twin model of the balancing hardware. This digital model is used as the environment during policy training. AI for Control.

35. A Curve Information Processing Method, Device, Storage Medium, and Detection Equipment (♠)

-  Peng Liu, Lin Sun, [Weijia Lu](#) and Tong Ma

 2023  CN 115631139 A

[Link](#)

About: AI for Manufacturing.
36. A Target Detection Method, Device, Storage Medium, Sensor, and Controller (♠♠♠)

 Peng Liu, [Weijia Lu](#), Lin Sun, Can Zhang and Tong Ma

 2023  CN 116452916 A

[Link](#)

About: Contrastive learning method for target detection. AI for Manufacturing.
37. A Target Detection Method, Machine Vision Device, Storage Medium, and Controller (♠♠♠)

 Peng Liu, Lin Sun, [Weijia Lu](#), Jie Zhang, Wei Shen, Yu Jin and Huan Ge

 2024  CN 117726855 A

[Link](#)

About: Semi-Supervised learning method for target detection. AI for Manufacturing.
38. A Product Testing Method, Data Management Method, Apparatus, Medium and Controller (♠♠)

 [Weijia Lu](#), Xiaodong Zhang, Can Zhang, Zhifei Yang, Xiaofeng Ma, Chuang Liu, Bingyang Li, Feng Wu, Xuzhou Zhang, Jing Ye, Yongyi Liu, Xichun Ke, Jianfei Zheng, Jie Bai and Chen Sheng

 2024  CN 118860853 A

[Link](#)

About: A test case generation tool utilizing signal matrix, IO configuration and large language model. LLM for Software Development.
39. A Method for Model Data Processing, a Simulation Apparatus, a Storage Medium, and a Testing System. (♠♠♠)

 Zhifei Yang, Xiaofeng Ma, [Weijia Lu](#), Xiaodong Zhang, Wangchen Lin, Ting Li, Fei Sun, Qiang Fang and Gang Xi

 2024  CN 118732531 B

[Link](#)

About: A method to establish digital twin model based on a new neural ODE structure. The method reported in this patent is mainly for time-series problem. AI for Plant Modeling.
40. Scenario-Aware Clustered Federated Learning for Vehicle Trajectory Prediction with Non-IID Data (♠)

 Liang Tao, Yangguang Cui, Xiaodong Zhang, Wenfeng Shen, [Weijia Lu](#)

 2024  Part D: Journal of Automobile Engineering (PartD)

[Link](#)

About: A vehicle trajectory model, federated learned from real vehicle data, with multi-head design and federated clustering. The corresponding method is protected in patent CN 116822619 A. Federated Learning in Vehicle.
41. A Comfortable and Robust DRL-based Car-following Policy Incorporating Lateral Information under Cut-in Scenarios (♠♠♠)

 Yifei Shen, Zhifei Yang, [Weijia Lu](#), Wenfeng Shen, Zhou Lei

 2024  in Proceedings of 35th IEEE Intelligent Vehicles Symposium (IV)

[Link](#)

About: A reinforcement learning policy to significant increase the safety. The vehicle trajectory model, federated learned from real vehicle data and reported in PartD 2024, provide critical lateral information. AI for Control.
42. Improving Generalization and Personalization in Long-Tailed Federated Learning via Classifier Retraining (♠♠)

 Yuhang Li, Liu Tong, Wenfeng Shen, Yangguang Cui, [Weijia Lu](#)

 2024  in Proceedings of 30th International European Conference on Parallel and Distributed Computing (Euro-par)

[Link](#)

About: A resampling strategy to address heterogenization issue of data distribution in a federated learning. Federated Learning in Vehicle.
43. Addressing Sim2Real Challenges in DRL-based Car-following Policies via Federated Reinforcement Learning and Augmented State Observations (♠)

 Yifei Shen, Zhifei Yang, [Weijia Lu](#), Wenfeng Shen, Xiaodong Zhang, Xiaofeng Ma

 2024  Proceedings of the 2024 International Conference on Intelligent Driving and Smart Transportation

[Link](#)
44. Hybrid Attention with Memory-based Conditional Refinement for Unified Intent Trajectory Prediction (♠♠♠)

 Jingjing Wang, [Weijia Lu](#), Wenfeng Shen, Yangguang Cui, Tong Liu, Xiaodong Zhang, Zhifei Yang, Guangtai Ding, Yifei Shen, Liang Tao

 2025  Transportation Research Record (TRR)

[Link](#)

About: A spatiotemporal network with semantic-search based training method. To improve the prediction of driver intention. AI for smart sensor.
45. FedSMU: Communication-Efficient and Generalization-Enhanced Federated Learning through Symbolic Model Updates (♠♠♠)

 Xinyi Lu, Hao Zhang, Chenglin Li, [Weijia Lu](#), ZHIFEI YANG, Wenrui Dai, xiaodong Zhang, Xiaofeng Ma, Can Zhang, Junni Zou, Hongkai Xiong

 2025  International Conference on Machine Learning (ICML) '25

[Link](#)

About: Alleviating data heterogeneity by symbolizing client model updates (normalizing parameter magnitude) and combining Lion optimizer's local-global operations reduces communication costs and enhances the global model's generalization. Federated Learning.
46. LBI-FL: Low-Bit Integerized Federated Learning with Temporally Dynamic Bit-Width Allocation (♠♠♠)

 Li Ding, Hao Zhang, Wenrui Dai, Chenglin Li, [Weijia Lu](#), ZHIFEI YANG, xiaodong Zhang, Xiaofeng Ma, Junni Zou, Hongkai Xiong

 2025  International Conference on Machine Learning (ICML) '25

[Link](#)

About: By using low-bit quantization and dynamic bit-width allocation via reinforcement learning, communication and computation costs are significantly reduced while maintaining high precision. Federated Learning.
47. Logical Consistency is Vital: Neural-Symbolic Information Retrieval for Negative-Constraint Queries (♠♠♠)

 Ganlin Xu, Zhoujia Zhang, Wangyi Mei, Jiaqing Liang, [Weijia Lu](#), xiaodong Zhang, ZHIFEI YANG, Xiaofeng Ma, Yanghua Xiao, Deqing Yang

 2025  The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)

[Link](#)

About: A neuro-symbolic info retrieval method using first-order logic to boost complex query retrieval. A simple but effective compensation for naive semantic-based RAG system. Information retrieval for LLM.

48. BPO: Revisiting Preference Modeling in Direct Preference Optimization(♠♠)

Lin Sun, Chuang Liu, Peng Liu, Bingyang Li, [Weijia Lu](#), Ning Wu

2025 [arXiv](#)

[Link](#)

About: Direct Preference Optimization (DPO) have emerged as a popular method for aligning Large Language Models (LLMs) with human preferences. While DPO effectively preserves the relative ordering between chosen and rejected responses through pairwise ranking losses, it often neglects absolute reward magnitudes. This oversight can decrease the likelihood of chosen responses and increase the risk of generating out-of-distribution responses, leading to poor performance. To address this issue, we propose Balanced Preference Optimization (BPO), a novel framework that dynamically balances the optimization of chosen and rejected responses through two key components: balanced reward margin and gap adaptor. Finetune of LLM.

49. FreePRM: Training Process Reward Models Without Ground Truth Process Labels (♠)

Lin Sun, Chuang Liu, Xiaofeng Ma, Tao Yang, [Weijia Lu](#), Ning Wu

2025 [arXiv](#)

[Link](#)

About: Recent advancements in Large Language Models (LLMs) have demonstrated that Process Reward Models (PRMs) play a crucial role in enhancing model performance. However, training PRMs typically requires step-level labels, either manually annotated or automatically generated, which can be costly and difficult to obtain at scale. To address this challenge, we introduce FreePRM, a weakly supervised framework for training PRMs without access to ground-truth step-level labels. Finetune of LLM.

50. A Design Assistance System (♠♠♠)

Yubing Wu, [Weijia Lu](#), Linming Li, Chenyu Bao, Chenbai Dai and Yangchao Liu

2025 [CN 202610881263.0](#)

[Link](#)

About: Ontology + GNN retrieval model.

51. ComLQ: Benchmarking Complex Logical Queries in Information Retrieval (♠♠)

Ganlin Xu, Zhitao Yin, Linghao Zhang, Jiaqing Liang, [Weijia Lu](#), xiaodong Zhang, ZHIFEI YANG, Sihang Jiang, Deqing Yang

2026 [The Fortieth AAAI Conference on Artificial Intelligence, 2026](#)

[Link](#)

About: Information retrieval (IR) systems play a critical role in navigating information overload across various applications. We propose a novel method leveraging large language models (LLMs) to construct a new IR dataset, which comprises 2,909 queries and 11,251 candidate passages. Our experimental results under zero-shot settings demonstrate existing retrieval models' limited performance on complex logical queries, especially on queries with negation, exposing their inferior capabilities of modeling exclusion. Information retrieval for LLM.

52. AgentPO: Enhancing Multi-Agent Collaboration via Reinforcement Learning (♠♠)

Lin Sun, Chuang Liu, Can Zhang, Yubin Wu, [Weijia Lu](#), Ning Wu

2026 [The Fourteenth International Conference on Learning Representations, 2026](#)

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About: Multi-Agent Systems (MAS) offer a powerful paradigm for solving complex problems through distributed reasoning and collaboration. However, their effectiveness is often hindered by the challenge of optimizing interactions among agents. To address this, we introduce AgentPO, a novel framework that directly optimizes agent collaboration. AgentPO employs reinforcement learning to train a specialized Collaborator agent, which refines its interaction policy to enhance overall system performance within a fixed multi-agent topology. We evaluated AgentPO on multiple mathematical reasoning tasks, where it consistently outperformed strong baselines. With Llama-3.2-3B-Instruct as the actor model, AgentPO achieves accuracy improvements of +1.8% and +7.2% over strong baselines like Role Assignment and EvoAgent, respectively. When using the larger Llama-3.1-8B-Instruct model, these gains increase to +5.6% and +11.3%. Crucially, AgentPO achieves these results with remarkable efficiency: it requires only 500 training samples and operates at just 7.8% of EvoAgent's inference cost, highlighting its superior scalability and practicality.

53. ProSAR: Prototype-Guided Semantic Augmentation and Refinement for Time Series Contrastive Learning(♠♠♠)

Caiyi Yang, Chenglin Li, Hao Zhang, [Weijia Lu](#), ZHIFEI YANG, Wenrui Dai, xiaodong Zhang, Xiaofeng Ma, Can Zhang, Junni Zou, Hongkai Xiong

2026 [International Conference on Machine Learning \(ICML\) '26](#)

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About: The paper ingeniously adapts traditional temporal-frequency feature analysis methods for prototype matching and data augmentation pipelines, pairing them with learnable prototypes and contrastive learning to ultimately achieve state-of-the-art results on five multivariate time series forecasting datasets as well as the UEA time series classification benchmark.

54. Learning to Think in Blocks: A Prior-Guided Reinforcement Learning Framework for Deep Research (♠♠)

Wangyi Mei, Ganlin Xu, Zhoujia Zhang, Jiaqing Liang, [Weijia Lu](#), Jinhui Huang, Chen Yang, Xiaodong Zhang, ZHIFEI YANG, Xiaofeng Ma, Deqing Yang

2026 [The 2026 Conference on Empirical Methods in Natural Language Processing, 2026](#)

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